QUALITY, RELIABILITY AND MAINTENANCE ME327E

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Evaluating Sampling Plans

Evaluating sampling plans involves analyzing how well a given plan performs, especially in the context of quality control. Several key measures are used to assess the effectiveness and efficiency of sampling plans.

Operating Characteristic (OC) Curve

- ***Definition***: The OC curve plots the probability of accepting a lot (P_a) versus the proportion nonconforming in the lot (p).
- ***Purpose***: It visually shows the discriminatory power of the plan—how well it can distinguish between good and poor quality lots.
- Steeper OC curves (more "S-shaped") indicate greater discriminatory power.
- A plan should ideally accept lots with low nonconformance most of the time and rarely accept lots with high nonconformance

Average Outgoing Quality (AOQ)

Definition: AOQ is the average quality level of lots *leaving* the inspection station, assuming rejected lots undergo 100% inspection (rectification).

Calculation: $AOQ = P_a \times p \times \frac{(N-n)}{N}$

- N: Lot size
- n: Sample size
- p: Incoming proportion nonconforming
- P_a: Probability lot is accepted

Interpretation: AOQ gives the long-term average nonconformance in outgoing lots after applying the sampling plan and rectification.

Average Outgoing Quality Limit (AOQL)

- **Definition:** AOQL is the *maximum* possible value for the AOQ across all levels of incoming quality.
- **Purpose:** It represents the worst average quality that can leave the inspection station, assuming rectification.
  - Gives a sense of risk protection: "the outgoing quality won't exceed AOQL."

Example

Construct an AOQ curve for a single sampling plan where the lot size is 2000, the sample size is 50, and the acceptance number is 2.

$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Lot Acceptance Probabilities for Different Values of Proportion Nonconforming for the Sampling Plan N = 2000, n = 50, c = 2

Proportion Nonconforming, p	np	Probability of Lot Acceptance, P_a	Proportion Nonconforming, p	np	Probability of Lot Acceptance, P_a
0.0	0.0	1.000	0.08	4.00	0.238
0.005	0.25	0.997	0.09	4.50	0.174
0.01	0.50	0.986	0.10	5.00	0.125
0.02	1.00	0.920	0.11	5.50	0.088
0.03	1.50	0.809	0.12	6.00	0.062
0.04	2.00	0.677	0.13	6.50	0.043
0.05	2.50	0.544	0.14	7.00	0.030
0.06	3.00	0.423	0.15	7.50	0.020
0.07	3.50	0.321			

Average Total Inspection (ATI)

- **Definition:** ATI is the average number of items inspected (including both the sample and any additional inspection for rejected lots) per lot.
- **Calculation for single sampling:**

$$ATI = n + (1 - P_a) \times (N - n)$$

  - When a lot is accepted, only the sample is inspected.
  - When rejected, the remainder is 100% inspected.
- **Use:** Helps estimate inspection costs for a sampling plan.

Exercise:

Construct the ATI curve for the sapling plan where $N = 2000$, $n = 50, c = 2$.

Average Sample Number (ASN)

- **Definition:** The average number of samples required to reach a decision, particularly for double and multiple sampling plans.
  - For single sampling:

$$ASN = n$$

  - For double sampling:

$$ASN = n_1 \cdot P_1 + (n_1 + n_2) \cdot (1 - P_1)$$

    - n_1: First sample size, n_2: Second sample size
    - P_1: Probability of decision after the first sample
- **Purpose:** Lower ASN means fewer inspections on average, reducing costs and time.

Exercise:

For the double sampling plan $N = 3000$, $n_1 = 40$, $c_1 = 1$, $r_1 = 4$, $n_2 = 80$, $c_2 = 3$, $r_2 = 4$, find the ASN for batches with proportion nonconforming of 0.02, assuming no curtailment. **Ans: 54.56**

Bayes Rule and Decision Making Based on Samples

Bayes Rule and decision making based on samples refers to using *Bayesian probability* theory to update knowledge or beliefs about an unknown parameter in quality control, by combining prior information and sample data.

Bayes Rule Concept

- **Prior Probabilities**: Before collecting any sample data, there is a "prior" belief or probability about possible states or values (e.g., which vendor supplied a part, expected demand proportion).
- **Sample Information**: Upon inspecting a sample, certain outcomes are observed (e.g., a part is defective).
- **Posterior Probabilities**: Bayes Rule is then used to update the prior probabilities by incorporating the likelihood of observing the sample outcome, to get new "posterior" probabilities for each possible state.

The mathematical formula described is:

$$P(B_i|A) = \frac{P(A|B_i) \cdot P(B_i)}{\sum_{i=1}^{m} P(A|B_i) \cdot P(B_i)}$$

where $P(A|B_i)$ is the probability of sample outcome A given B_i, and $P(B_i|A)$ is the posterior probability after seeing the outcome A.

Exercise

Three vendors, B1, B2, and B3, provide microchips for a semiconductor manufacturer. Based on each vendor's historical data, the probabilities of vendor B1, B2, and B3 producing a defective microchip are 0.01, 0.03, and 0.05, respectively. Since the parts all look alike and have no company identification, it is not possible to identify the vendor they came from. Assume that 30% of the parts are provided by vendor B1, with the corresponding percentages for vendors B2 and B3 being 50% and 20%, respectively. Upon inspection of a part, it is found to be nonconforming. What is the probability that it was produced by vendor B1? By vendor B2? By vendor B3?

$$\boldsymbol{P(B_1|A) = 0.107, P(B_2|A) = 0.536\ \&\ P(B_3|A) = 0.357}$$

Quality Control Application

- **Vendor Identification Example**: If there are multiple vendors with different defect rates and proportions of supplied parts, and a defective item is found, Bayes rule helps calculate the likelihood that the item came from each vendor based on both prior production statistics and the defect rates.
- **Decision Updates**: As more sample results are obtained, the probability distribution for a parameter (such as defect rate or expected demand) is continually updated. This sequential learning allows better-informed decisions about process improvement, vendor selection, or product launch strategy.

Exercise

An electronics company is considering the addition of a new product to its line. It currently has 40,000 customers. Management estimates product development costs to be Rs. 12,00,00,000,which may be assumed to be the overhead costs. Variable manufacturing costs per unit are expected to be Rs. 80,000, along with variable selling expense of Rs. 10,000 per unit. Units will be produced on a demand basis. From a market survey, it is estimated that the new product could be priced at Rs. 1,50,000 per unit. Based on previous experience with similar products, market research has come up with estimates of the proportion of existing customers that may purchase the new product and the corresponding probabilities of such happening are shown in table. Should the company develop the new product?

Probability Distribution of Demand for a New Product

Demand Probability, p	Probability
0.02	0.2
0.04	0.4
0.06	0.3
0.08	0.1